

CSE 3175: ARTIFICIAL NEURAL NETWORKS [3 0 0 3]

Models of a Neuron, Architectures, Learning, Perceptron, Convergence Theorem, Multilayer Perceptron, Back-Propagation, XOR Problem, Multilayer Perceptron Applications. Cover's Theorem, Interpolation Problem, Radial-Basis-Function Networks, K-Means Clustering. Feature Mapping Models, Self-Organizing Map, Properties of the Feature Map, Case Study. Recurrent Network Architectures, Computational Power, Back Propagation Through Time, Real-Time Recurrent Learning. Hopfield Network, Associative Memories, Storage Capacity of Memories.

References:

1. Simon Haykin, Neural Networks and Learning Machines, 3rd ed, Pearson, Prentice Hall, 2009
2. Ivan Nunes da Silva, Danilo Hernane Spatti, Rogerio Andrade Flauzino, Luisa Helena Bartocci Liboni, & Silas Franco dos Reis Alves, Artificial Neural Networks: A Practical Course, Springer International Publishing, 2017
3. John Paul Mueller & Luca Massaron, Deep Learning For Dummies, Wiley & Sons, Inc., 2019
4. Daniel Graupe, Principles of Artificial Neural Networks, 3rd ed, World Scientific Publishing, 2013
5. Eugene Charniak, Introduction to Deep Learning, MIT Press, 2018

ARTIFICIAL NEURAL NETWORK [Revised Credit System] (Effective from the academic year 2021-2022) SEMESTER - V			
Subject Code	CSE 3175	IA Marks	50
Number of Lecture Hours/Week	03	Exam Marks	50
Total Number of Lecture Hours	36	Exam Hours	03
CREDITS – 03			
Course objectives: This course will enable students to • To understand different neural network architectures. • To train the neural networks. • To design a neural network system for a given problem. • To appreciate Kernel-based methods			
Module -1			Teaching Hours

INTRODUCTION: What is a Neural Network? The Human Brain, Models of a Neuron, Network Architectures, Learning Processes, Learning Tasks. Text Book 1: Selected topics from Introduction	5 Hours
Module -2	
ROSENBLATT’S PERCEPTRON & MULTILAYER PERCEPTRON: Introduction to Rosenblatt’s Perceptron, Perceptron, The Perceptron Convergence Theorem, Introduction to Multilayer Perceptron, Some Preliminaries, Batch Learning and On-Line Learning, The Back-Propagation Algorithm, XOR Problem, Heuristics for Making the Back-Propagation Algorithm Perform Better, Multilayer Perceptron Applications. Text Book 1: 1.1-1.3, 4.1-4.6. Text Book 2: 5.4	12 Hours
Module – 3	
KERNEL METHODS & RADIAL-BASIS FUNCTION NETWORKS: Introduction, Cover’s Theorem on the Separability of Patterns, The Interpolation Problem, Radial-Basis-Function Networks, K-Means Clustering, Recursive Least-Squares Estimation of the Weight Vector, Hybrid Learning Procedure for RBF Networks. Text Book 1: 5.1-5.7	5 Hours
Module – 4	
SELF-ORGANIZING MAPS: Introduction, Two Basic Feature-Mapping Models, Self-Organizing Map, Properties of the Feature Map, Case Study: Pattern Identification. Text Book 1: 9.1-9.4, Text Book 2: Chapter 15	4 Hours
Module- 5	

DYNAMICALLY DRIVEN RECURRENT NETWORKS: Introduction, Recurrent Network Architectures, Universal Approximation Theorem, Controllability and Observability, Computational Power of Recurrent Networks, Learning Algorithms, Back Propagation Through Time, Real-Time Recurrent Learning. Text Book 1: 15.1-15.8	6 Hours
Module -6	
RECURRENT HOPFIELD NETWORKS: Introduction, Operating Principles of the Hopfield Network, Stability Conditions of the Hopfield Network, Associative Memories, Outer Product Method, Pseudoinverse Matrix Method, Storage Capacity of Memories, Design Aspects of the Hopfield Network. Text Book 2: 7.1-7.5	4 Hours
Course outcomes:	
After studying this course, students will be able: 1. To understand the different types of learning, activation functions, and simple Perceptron. 2. To analyze MLP and RBF networks. 3. To model and design SOM systems for a given problem. 4. To recognize the need for recurrent and Hopfield networks. 5. To understand back-propagation method for neural networks.	
Text Books: 1. Simon Haykin, <i>Neural Networks and Learning Machines</i>, 3rd ed, Pearson, Prentice Hall, 2009 2. Ivan Nunes da Silva, Danilo Hernane Spatti, Rogerio Andrade Flauzino, Luisa Helena Bartocci Liboni, & Silas Franco dos Reis Alves, <i>Artificial Neural Networks: A Practical Course</i>, Springer International Publishing, 2017 3. John Paul Mueller & Luca Massaron, <i>Deep Learning For Dummies</i>, Wiley & Sons, Inc., 2019	
Reference Books: 1 Daniel Graupe, <i>Principles of Artificial Neural Networks</i>, 3rd ed, World Scientific Publishing, 2013 2 Eugene Charniak, <i>Introduction to Deep Learning</i>, MIT Press, 2018	